

# The Magic Thermometer

## Student sheet

Name: \_\_\_\_\_

Date: \_\_\_\_\_

### 1. What I see on the screen

On the screen you can see the cards **Temperature** (*in degrees Celsius*) and **Humidity** (*in percent*), and below them their chart. You start with **Start**; you move on with the green button at the bottom right, you go back with **Back**, and the last green button reads **Finish**.

**The sensor is slow.** After you move the box, **wait 60–90 seconds** before you write the number down. The box stays on the table, not in your hand, and nobody breathes on it.

### 2. What I do on my own

**First I predict. I do not change my prediction after I measure.**

1. Is the whole room the same temperature? ☐ yes, the same everywhere ☐ no, it is different from place to place

I think the table is about \_\_\_\_\_ °C. Put an arrow on the line, where you think the room is today:



2. **Heat source** (circle one): computer   radiator   lamp   warm water. Beside it I think it will be:

☐ warmer   ☐ cooler   ☐ the same   by about \_\_\_\_\_ °C

because \_\_\_\_\_

**While I wait, I draw the map of the room.** One dot for every place I will measure, with its name beside it. After I measure, I write the temperature next to each dot: blue if it is cool, red if it is warm.

| Place                   | Temperature<br>(°C) | Humidity<br>(%) | Circle one        |
|-------------------------|---------------------|-----------------|-------------------|
| On the table (the room) |                     |                 | cool / warm / hot |
| Beside the heat source  |                     |                 | cool / warm / hot |
| On the floor            |                     |                 | cool / warm / hot |
| On top of the cupboard  |                     |                 | cool / warm / hot |

When the temperature went up, the humidity went: ☐  $\uparrow$    ☐  $\downarrow$    ☐  $\rightarrow$

I predicted that \_\_\_\_\_

The data show that \_\_\_\_\_

The difference is explained by \_\_\_\_\_